

End of Topic Test – Chapter 6: Confidence Intervals – Test 2

Mark Scheme

Total: 48 marks

1. Confidence interval for a mean

$n = 20$, $\bar{x} = 215.6$, $s = 18.4$.

- (a) M1 degrees of freedom $= n - 1 = 19$ identified. M1 critical value $t_{0.025,19} = 2.093$ used. M1 standard error $= s/\sqrt{n} = 18.4/\sqrt{20} = 4.115$. A1 margin of error $= 2.093 \times 4.115 = 8.6$ (3 s.f.). A1 CI $= (215.6 - 8.6, 215.6 + 8.6) = (207.0, 224.2)$ ms (1 d.p.). [5]
- (b) B1 we are 95% confident the true (population) mean reaction time of athletes lies between 207.0 ms and 224.2 ms. B1 the “95% confidence” refers to the method: if many samples were taken and a CI constructed from each in the same way, about 95% of those intervals would contain the true population mean. B1 stated clearly in context (athletes/reaction time). [3]
- (c) B2 (or B1+B1) the population standard deviation σ is unknown, so the sample standard deviation s is used as an estimate; this extra uncertainty (especially with a relatively small sample, $n = 20$) is accounted for by the t -distribution’s heavier tails compared with the Normal z -distribution. [2]

2. Confidence interval for a proportion

$n = 500$, $x = 310$, $\hat{p} = 310/500 = 0.62$.

- (a) M1 $\hat{p} = 0.62$. M1 SE $= \sqrt{0.62 \times 0.38/500} = 0.0217$. M1 margin $= 1.96 \times 0.0217 = 0.0425$. A1 CI $= (0.5775, 0.6625)$, i.e. (57.7%, 66.3%) to 1 d.p. [4]
- (b) B1 compare the interval to 55%: the entire interval (57.7% to 66.3%) lies above 55%. B1 conclusion: the sample data supports the retailer’s claim that at least 55% of customers prefer the new design. B1 caveat: the lower bound (57.7%) is not very far above 55%, so the claim is supported but the margin is not especially large. [3]
- (c) B1 condition: $n\hat{p} \geq 5$ and $n(1 - \hat{p}) \geq 5$ (needed for the Normal approximation to be valid). B1 verified: $n\hat{p} = 500 \times 0.62 = 310 \geq 5$ and $n(1 - \hat{p}) = 500 \times 0.38 = 190 \geq 5$, so the condition holds. [2]

3. Sample size for a desired margin of error

$E \leq 1.5$, $\sigma \approx 6.0$, 95% confidence ($z = 1.96$).

- (a) M1 formula $n \geq \left(\frac{z\sigma}{E}\right)^2$. M1 substitute: $n \geq \left(\frac{1.96 \times 6.0}{1.5}\right)^2 = (7.84)^2$. A1 $n \geq 61.47$. A1 round up to the next whole number: $n = 62$. [4]
- (b) B1 the critical value z decreases for 90% confidence (from 1.96 to 1.645). B1 since $n \propto z^2$, a smaller z means a smaller required sample size is needed to achieve the same margin of error at the lower confidence level. [2]

4. Critical values and confidence levels

- (a) B1 $z_{0.05} = 1.645$ is the value such that 5% of the standard Normal distribution lies in each tail beyond ± 1.645 , leaving 90% in the middle. B1 the confidence interval is constructed as $\bar{x} \pm 1.645 \times \text{SE}$, using 1.645 as the multiplier of the standard error. B1 clear, complete explanation linking the shaded central region to the 90% confidence level. [3]
- (b) B1 critical value for 99% confidence is $z = 2.576$ (2.58). B1 since this is larger than 1.645, the margin of error (and so the width of the interval) is greater – the 99% interval is considerably wider than the 90% interval, all else being equal. [2]

5. Interpretation and misconceptions

- (a) B3 (or B1 partial credit) correct non-technical interpretation, e.g. “We are 99% confident that the true population mean lies between 45.3 and 52.7” – i.e. if this method of estimation were repeated many times, about 99% of the resulting intervals would contain the true mean. [3]
- (b) B1 the population mean is a fixed (though unknown) number, not a random variable, so for this particular interval it either does or does not contain the true mean – there is no “99% probability” attached to this one interval. B1 the 99% refers to the long-run performance of the method across repeated samples, not a probability statement about this specific interval. B1 clear overall explanation of the distinction. [3]
- (c) B2 (or B1+B1) valid reason, e.g. a narrower interval gives a more precise estimate of the true value, which may be important when a precise decision or comparison needs to be made, even though it carries a lower level of confidence (there is a trade-off between precision and confidence). [2]

6. Applying confidence intervals to decision making

$n = 40$, $\bar{x} = 18.4$, $s = 2.4\text{g}$. Target maximum = 20g.

- (a) M1 $\text{SE} = 2.4/\sqrt{40} = 0.3795$. M1 margin = $1.96 \times 0.3795 = 0.744$. A1 CI = $(18.4 - 0.744, 18.4 + 0.744)$. A1 = (17.66, 19.14) grams (2 d.p.). [4]
- (b) B1 compare interval to target: the entire interval (17.66 to 19.14) lies below 20g. B1 since 20g is not contained in (and lies entirely above) the interval, this is evidence the true mean sugar content under the new recipe is below the target. B1 clear conclusion in context (the new recipe appears to have reduced the mean sugar content below 20g). [3]
- (c) B1 criterion stated: upper bound of the 95% CI must not exceed 19.5g. B1 compare: the calculated upper bound is 19.14g, which is $\leq 19.5\text{g}$. B1 conclusion: the criterion is met, so the recipe qualifies for the “reduced sugar” label. [3]